Analog Electronics II: Op-amp & 555 Timer

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EEE 311

Lecture # 7b

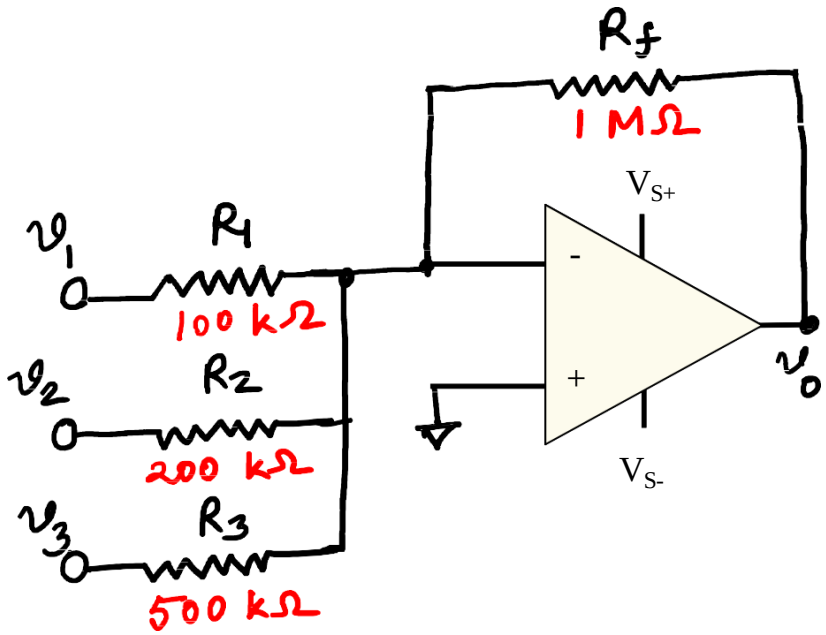
Practice Problems

Inverting Weighted Adder

R_f should be a common multiple of 10, 5, 2.

Design an op-amp-based circuit so that the output voltage v_o is related to the input voltages v_1, v_2, v_3 as follows:

$$v_o = -10v_1 - 5v_2 - 2v_3 = A_{cl_1} \cdot v_1 + A_{cl_2} \cdot v_2 + A_{cl_3} \cdot v_3$$



$$\Rightarrow v_o = \left(-\frac{R_f}{R_1}\right)v_1 + \left(-\frac{R_f}{R_2}\right)v_2 + \left(-\frac{R_f}{R_3}\right)v_3$$

$$A_{cl_1} = -\frac{R_f}{R_1} = -10 \Rightarrow R_1 = \frac{R_f}{10} \quad (1)$$

$$A_{cl_2} = -\frac{R_f}{R_2} = -5 \Rightarrow R_2 = \frac{R_f}{5} \quad (2)$$

$$A_{cl_3} = -\frac{R_f}{R_3} = -2 \Rightarrow R_3 = \frac{R_f}{2} \quad (3)$$

4 unknown variables: R_f, R_1, R_2, R_3

3 eq^{n}s

From eq^n (1) – (3),

Assume, $R_f = 1\ M\Omega$

$R_1 = 100\ k\Omega$

$R_2 = 200\ k\Omega$

$R_3 = 500\ k\Omega$

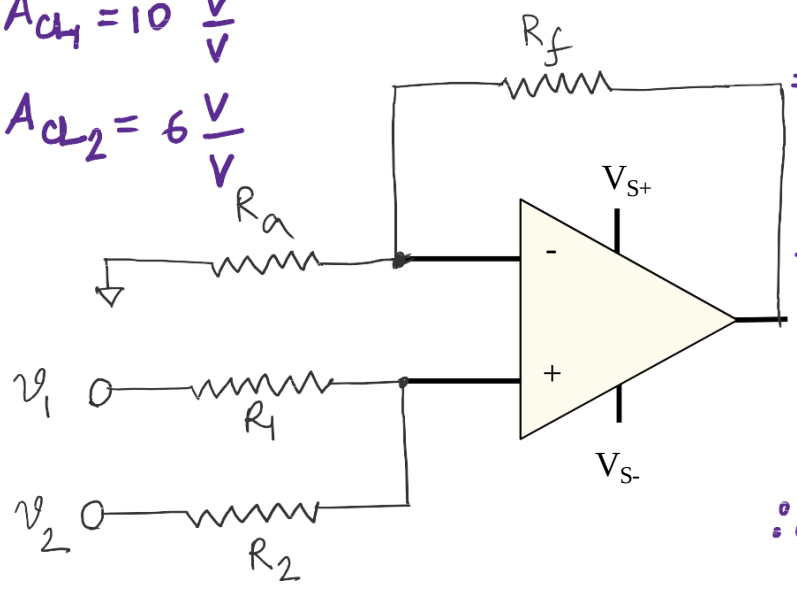
Non-inverting Weighted Adder

Design an op-amp-based weighted non-inverting adder circuit so that the output voltage as follows:

$v_o = 10v_1 + 6v_2 = A_{CL_1} \cdot v_1 + A_{CL_2} \cdot v_2 = \left\{\left(1+\frac{R_f}{R_a}\right)\cdot\frac{R_2}{R_1+R_2}\right\} v_1 + \left\{\left(1+\frac{R_f}{R_a}\right)\cdot\frac{R_1}{R_1+R_2}\right\} v_2$

$A_{CL_1} = 10 \frac{V}{V}$

$A_{CL_2} = 6 \frac{V}{V}$



$= \left(1+\frac{R_f}{R_a}\right)\cdot\frac{1}{R_1+R_2}\cdot\left\{R_2 \cdot v_1 + R_1 \cdot v_2\right\}$

$= A_x \cdot \left\{R_2 \cdot v_1 + R_1 \cdot v_2\right\}$

$\left[\text{Assuming, } A_x = \left(1+\frac{R_f}{R_a}\right)\cdot\frac{1}{R_1+R_2}\right]$

$\therefore A_{CL_1} = A_x \cdot R_2 = 10 \frac{V}{V} \Rightarrow R_2 \propto 10$

$A_{CL_2} = A_x \cdot R_1 = 6 \frac{V}{V} \Rightarrow R_1 \propto 6$

Hence, we should start the design with assuming R_1 and R_2. $\therefore R_2 \propto A_{CL_1}$ and $R_1 \propto A_{CL_2}$

If we assume R_f and R_a, R_1 and R_2 will not be proportional to A_{CL_2} and A_{CL_1}, respectively.

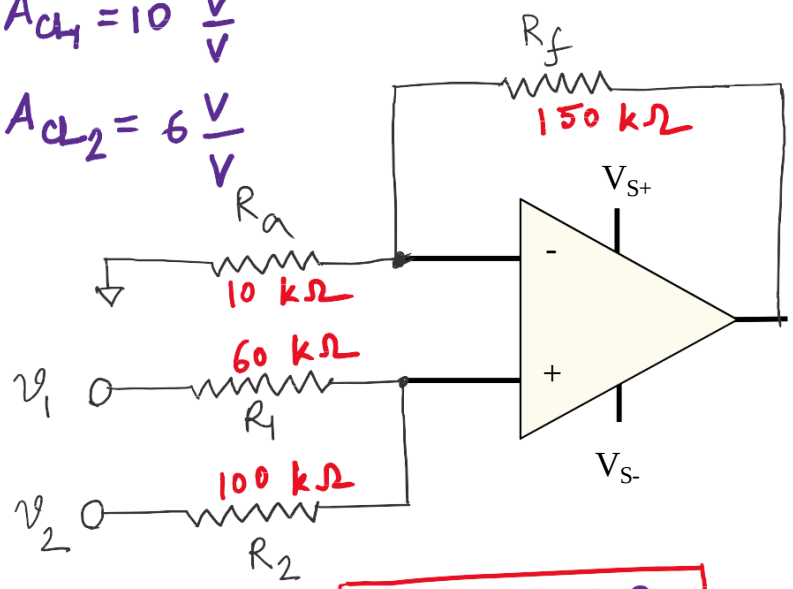
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$A_{CL_1} = 10\ \frac{V}{V}$

$A_{CL_2} = 6\ \frac{V}{V}$



Let's assume,

$R_2 = 100\ k\Omega$

$R_1 = 60\ k\Omega$

Assume, $R_a = 10\ k\Omega$

$= \left\{\left(1+\frac{R_f}{R_a}\right)\cdot\frac{100}{60+100}\right\}\cdot v_1 + \left\{\left(1+\frac{R_f}{R_a}\right)\cdot\frac{60}{60+100}\right\}v_2$

$= \left\{\left(1+\frac{R_f}{R_a}\right)\times 0.625\right\}\cdot v_1 + \left\{\left(1+\frac{R_f}{R_a}\right)\times 0.375\right\}\cdot v_2$

Now, $\left(1+\frac{R_f}{R_a}\right)\times 0.625 = A_{CL_1} = 10\ \frac{V}{V}$

$\Rightarrow 1+\frac{R_f}{R_a} = \frac{10}{0.625} = 16$

$\Rightarrow \frac{R_f}{R_a} = 16-1 = 15 \Rightarrow R_f = 15R_a \cdots\cdots (1)$

Then, $\left(1+\frac{R_f}{R_a}\right)\times 0.375 = 6 \Rightarrow 1+\frac{R_f}{R_a} = 16$

$\Rightarrow R_f = 15R_a \cdots\cdots (11)$

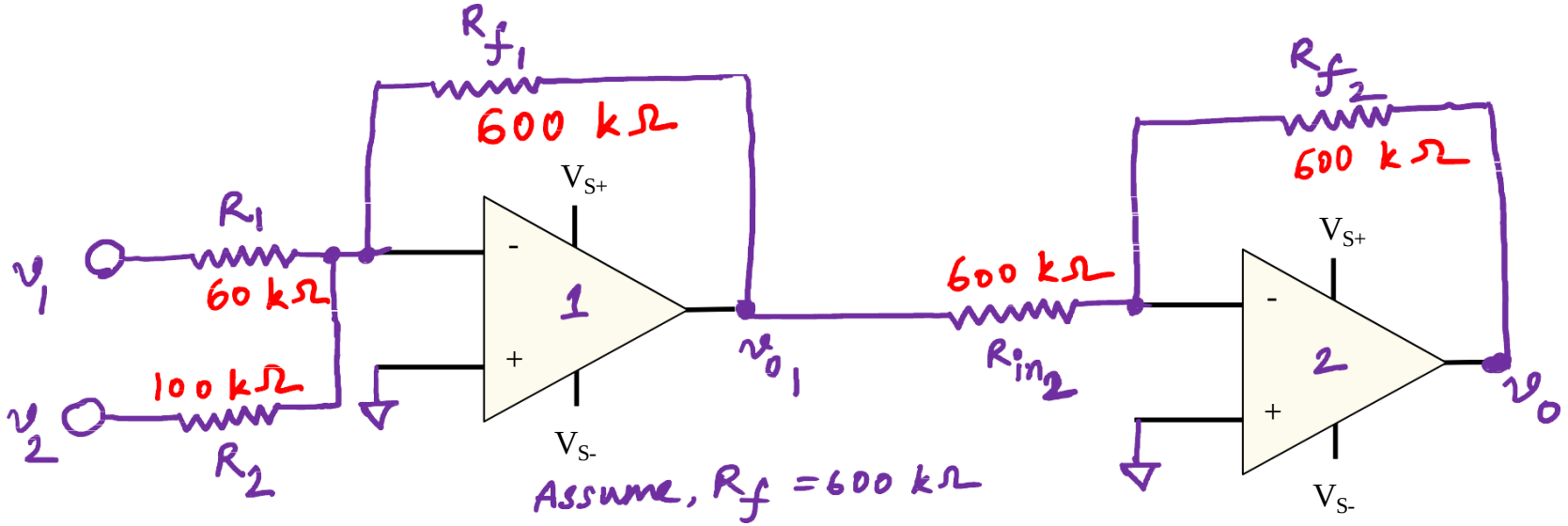
$\therefore R_f = 15\times 10\ k\Omega \Rightarrow \boxed{R_f = 150\ k\Omega}$

Cascaded Inverting Adders & Amplifiers

Design an op-amp-based circuit so that the output voltage v_o is related to the input voltages v_1, v_2 as follows:

$v_o = 10v_1 + 6v_2 = (-10v_1 - 6v_2)\cdot(-1)$

-ve x -ve = +ve



Assume, $R_f = 600\ k\Omega$

$A_{cl_1} = +10$

$-\frac{R_f}{R_1} = -10 \Rightarrow R_1 = \frac{R_f}{10}$

$A_{cl_2} = +6$

$-\frac{R_f}{R_2} = -6 \Rightarrow R_2 = \frac{R_f}{6}$

$-\frac{R_{f_2}}{R_{in_2}} = -1 \Rightarrow R_{f_2} = R_{in_2}$

Assume,

$R_{f_2} = 600\ k\Omega = R_{in_2}$

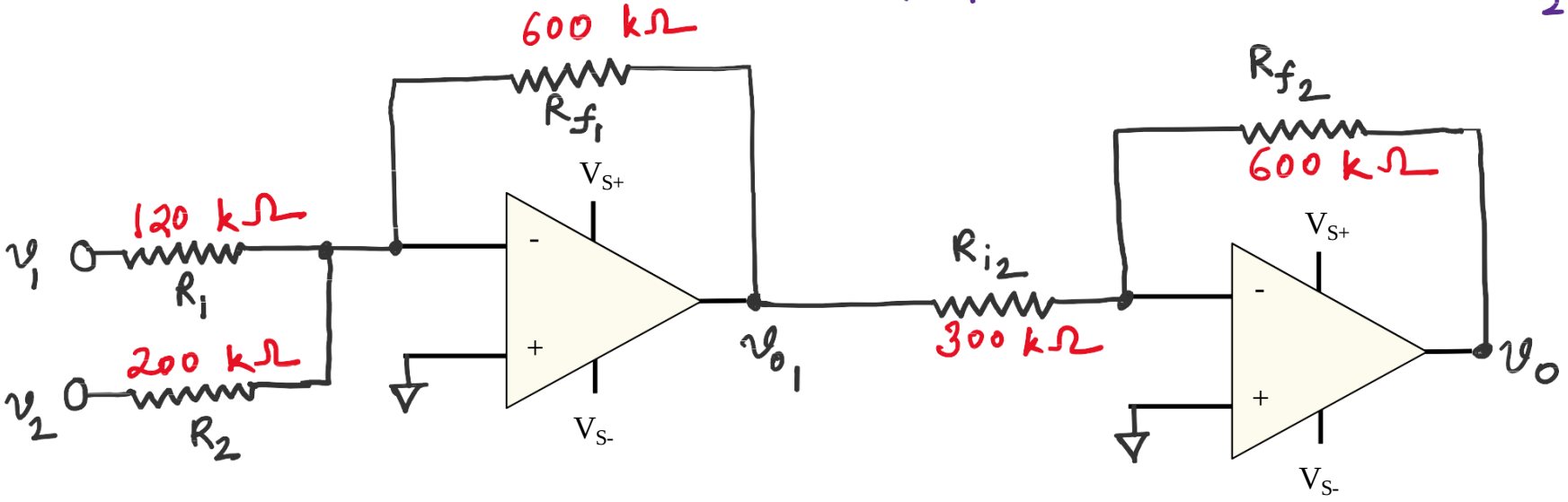
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Cascaded Inverting Adders & Amplifiers

Design an op-amp-based circuit so that the output voltage v_o is related to the input voltages v_1, v_2 as follows:

Problem 29. Alternative solution (2)

$$v_o = 10v_1 + 6v_2 = \underbrace{(-5v_1 - 3v_2)}_{\text{inverting}}\underbrace{(-2)}_{\text{inv. amp.}} = \left\{\left(-\frac{R_{f_1}}{R_1}\right)v_1 + \left(-\frac{R_{f_1}}{R_2}\right)\cdot v_2\right\}\cdot\left(-\frac{R_{f_2}}{R_{i_2}}\right)$$



$A_{CL_1} = +10 \ \frac{V}{V}$

Let's assume, $R_{f_1} = 600\ k\Omega$ and $R_{f_2} = 600\ k\Omega$

$+\frac{R_{f_1}}{R_1} = +5 \Rightarrow R_1 = \frac{R_{f_1}}{5} \Rightarrow R_1 = 120\ k\Omega$

$+\frac{R_{f_1}}{R_2} = +3 \Rightarrow R_2 = \frac{R_{f_1}}{3} \Rightarrow R_2 = 200\ k\Omega$

$-\frac{R_{f_2}}{R_{i_2}} = -2 \Rightarrow R_{i_2} = \frac{R_{f_2}}{2}$

$\therefore R_{i_2} = 300\ k\Omega$

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Cascaded Differential Amplifier

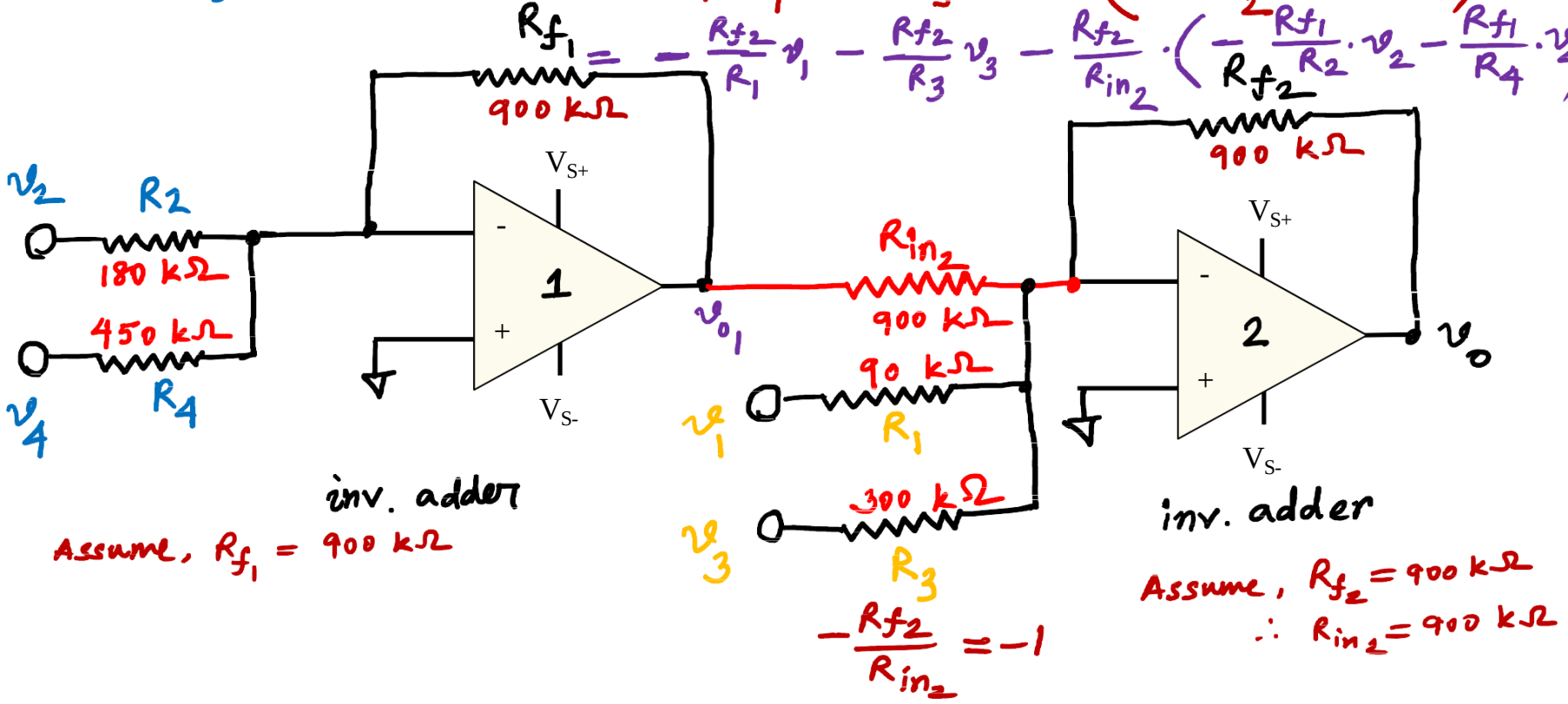
Design an op-amp-based circuit so that the output voltage v_o v_1, v_2, v_3, v_4 as follows:

$$v_o = -10v_1 + 5v_2 - 3v_3 + 2v_4$$

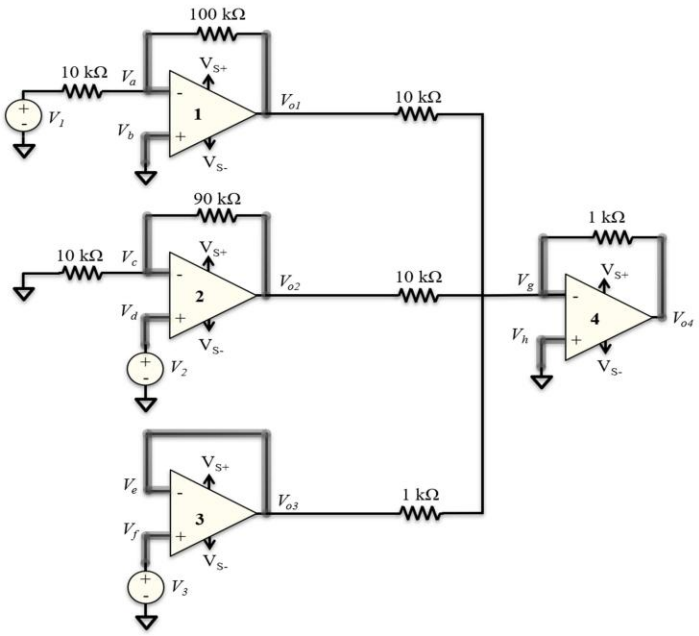
Do all calculation as shown in the above solutions.

$$= -10v_1 - 3v_3 - 1\left(-5v_2 - 2v_4\right)$$

$$= -\frac{R_{f2}}{R_1}v_1 - \frac{R_{f2}}{R_3}v_3 - \frac{R_{f2}}{R_{in_2}}\cdot\left(-\frac{R_{f1}}{R_2}\cdot v_2 - \frac{R_{f1}}{R_4}\cdot v_4\right)$$



Question. In the following circuit, calculate V_{o1}, V_{o2}, V_{o3}, and V_{o4}.



Question. Design an adder / subtractor circuit to achieve the following output voltage expression:

$$V_0 = V_1 + 2V_2 + \frac{1}{2} V_3 - V_4$$

Assume ideal op-amp with bias voltages of ±15 V. All input voltages are DC. You may use multiple op-amps.

Make sure that there is no chance of signal strength loss at any point of the circuit.

Question. Design an adder / subtractor circuit to achieve the following output voltage expression:

$$V_0 = V_1 + V_2 - 2V_3 - \frac{1}{2}V_4$$

Assume ideal op-amp with bias voltages of ±15 V. All input voltages are DC. You may use multiple op-amps.

Make sure that there is no chance of signal strength loss at any point of the circuit.

Question. Analyze the circuit using KVL, KCL, and Ohm's law. Then, calculate all branch currents and node voltages for the following circuit

$I_a =$

$I_f =$

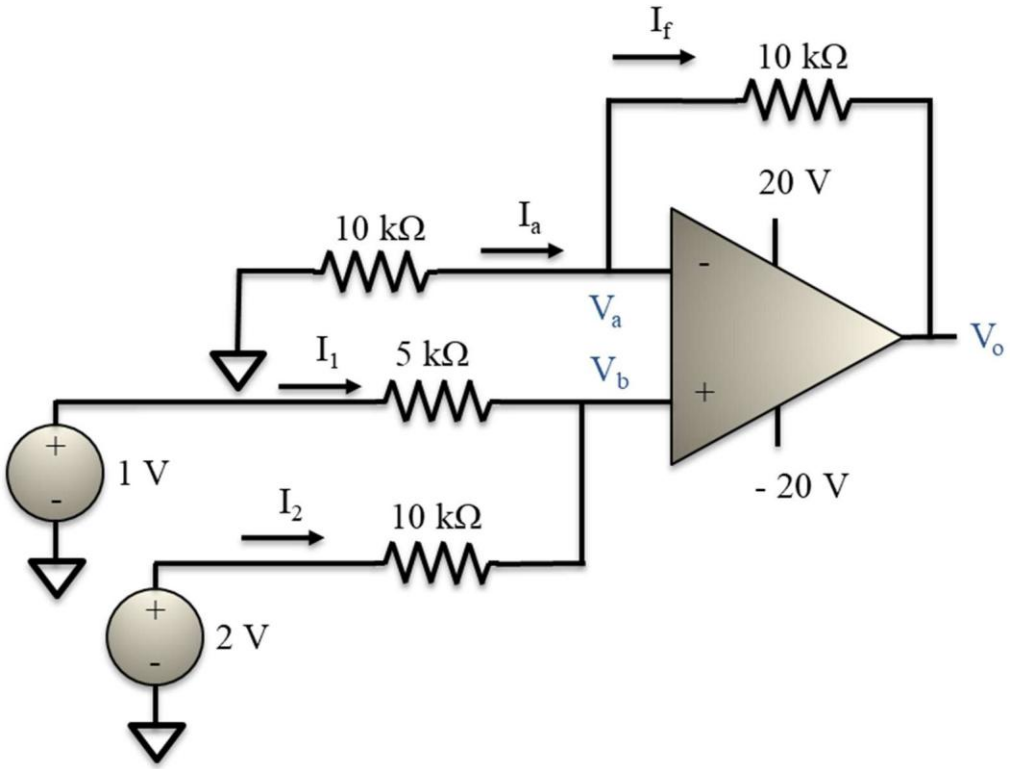
$I_1 =$

$I_2 =$

$V_a =$

$V_b =$

$V_0 =$



Question. Express V_o as a function of V_1, V_2, and V_3 for the following circuit.

